

Inorganic Chemistry

1. Introduction

Inorganic chemistry is the branch of chemistry that deals with the properties, structures, reactions, and applications of elements and their compounds, except most carbon compounds. It includes s-block, p-block, d-block, f-block, coordination compounds, organometallics, bioinorganic chemistry, and solid-state chemistry.

At the BSc level, inorganic chemistry is a core subject because it explains periodic trends, bonding, metal-ligand interactions, oxidation states, coordination geometry, magnetism, spectroscopy, and industrial and biological applications of metals.

2. Atomic structure and periodicity

Atomic structure provides the basis for understanding element properties and chemical bonding. Periodic trends arise from changes in effective nuclear charge, atomic size, ionization enthalpy, electron affinity, and electronegativity.

Key periodic trends

- Atomic radius decreases across a period and increases down a group.
- Ionization enthalpy generally increases across a period.
- Electronegativity increases across a period.
- Metallic character decreases across a period.

Effective nuclear charge explains many periodic changes because outer electrons feel different attraction from the nucleus depending on shielding.

3. Chemical bonding

Chemical bonding in inorganic chemistry includes ionic bonding, covalent bonding, coordinate bonding, metallic bonding, and advanced orbital-based theories.

Major theories

- Lewis theory.
- VSEPR theory.
- Valence bond theory.
- Molecular orbital theory.
- Crystal field theory.
- Ligand field theory.

4. Molecular orbital theory

MO theory explains bonding and antibonding orbitals formed by combination of atomic orbitals. It is especially important for diatomic molecules, bond order calculation, and magnetic behavior.

Key points

- Bond order = $\frac{N_b - N_a}{2}$.
- Greater bond order means greater stability and shorter bond length.
- Paramagnetism arises from unpaired electrons.
- Diamagnetism arises from paired electrons.

5. s-block elements

s-block chemistry includes Group 1 and Group 2 elements. These elements are highly electropositive and form mainly ionic compounds.

Group 1 elements

- Alkali metals.
- Soft, highly reactive, low ionization enthalpy.
- Form basic oxides and hydroxides.

Group 2 elements

- Alkaline earth metals.
- Less reactive than Group 1.
- Form oxides, hydroxides, and salts with characteristic trends.

6. Important s-block topics

Anomalous behavior of lithium and beryllium

Because of small size, high charge density, and polarizing power, lithium and beryllium show behavior different from the rest of their groups.

Diagonal relationship

Lithium resembles magnesium and beryllium resembles aluminum due to similar size and charge density ratios.

Hardness and solubility trends

Solubility and thermal stability of salts vary systematically down the groups.

7. p-block elements

p-block chemistry is vast and includes Groups 13 to 18. It covers covalency, oxidation states, inert pair effect, allotropy, oxides, hydrides, halides, and oxyacids.

General trends

- Increasing metallic to nonmetallic character across a period.
- Variable oxidation states.
- Multiple bonding in lighter elements.
- Inert pair effect in heavier elements.

8. Group 13 chemistry

Group 13 includes boron, aluminum, gallium, indium, and thallium. Boron is nonmetallic and forms electron-deficient compounds, while heavier members show increasing metallic character.

Topics

- Boranes.
- Diborane structure.
- Electron-deficient bonding.
- Borax and boric acid.
- Aluminum compounds and amphotericism.

9. Group 14 chemistry

Group 14 includes carbon, silicon, germanium, tin, and lead. The group exhibits catenation, allotropy, and variable oxidation states.

Important points

- Carbon shows strong catenation.
- Silicon chemistry is dominated by silicates and silicones.
- Tin and lead show inert pair effect.

10. Group 15 chemistry

Group 15 includes nitrogen, phosphorus, arsenic, antimony, and bismuth. The chemistry is dominated by hydrides, oxides, oxyacids, and catenation in phosphorus.

Important topics

- Ammonia and ammonium salts.
- Phosphorus allotropes.
- Phosphine.
- Oxoacids of phosphorus.
- Inert pair effect in heavier members.

11. Group 16 chemistry

Group 16 includes oxygen, sulfur, selenium, tellurium, and polonium. Their chemistry includes hydrides, oxides, oxyacids, and allotropy.

Topics

- Water and hydrogen peroxide.
- Sulfur allotropes.
- Sulfur oxides.
- Sulfuric acid and sulfurous acid.
- Oxidation states and catenation.

12. Group 17 chemistry

Halogens are highly electronegative and reactive nonmetals. Their chemistry includes interhalogens, oxyacids, and halides.

Topics

- Fluorine anomaly.
- Oxidizing power.
- Hydrogen halides.
- Halogen oxides.
- Interhalogen compounds.

13. Group 18 chemistry

Noble gases were once considered inert, but many compounds of xenon and krypton are now known. Their chemistry is important in understanding unusual oxidation states and bonding under special conditions.

14. d-block elements

d-block chemistry focuses on transition metals, which show variable oxidation states, complex formation, colored ions, catalytic activity, and magnetic properties.

Reasons for transition behavior

- Partially filled d orbitals.
- Similar energies of $(n - 1)d$ and ns orbitals.
- Variable oxidation states.

Key properties

- Colored compounds.
- Magnetic behavior.
- Catalysis.
- Complex formation.
- Alloy formation.

15. Coordination compounds

Coordination chemistry is one of the most important parts of inorganic chemistry. It studies complexes formed by coordination of ligands to a central metal atom or ion.

Core ideas

- Coordination number.
- Ligand types.
- Chelation.
- Stability constants.
- Isomerism.
- Nomenclature.

Isomerism

- Ionization isomerism.
- Hydrate isomerism.
- Linkage isomerism.
- Coordination isomerism.
- Geometrical isomerism.
- Optical isomerism.

16. Crystal field theory

Crystal field theory explains the splitting of d orbitals in the presence of ligands. It is essential for understanding color, magnetism, and stability of complexes.

Octahedral field

- t_{2g} and e_g splitting.
- CFSE calculation.
- High-spin and low-spin cases.

Tetrahedral field

- Smaller splitting than octahedral.
- Usually high spin.

Square planar field

- Important for d^8 complexes such as Pt(II) and Ni(II) in some cases.

17. Ligand field theory and MOT for complexes

Ligand field theory combines CFT with molecular orbital concepts. It explains sigma bonding, pi bonding, metal-ligand overlap, and electronic structure more completely than CFT alone.

18. Stability and reaction of complexes

Complex stability depends on charge, size, chelate effect, ligand field stabilization, and metal-ligand bonding. Reactions may proceed through associative or dissociative substitution mechanisms.

19. Organometallic chemistry

Organometallic chemistry deals with compounds containing metal-carbon bonds. It is a major branch of modern inorganic chemistry and underlies catalysis in industry.

Key topics

- 18-electron rule.
- σ - and π -bonding.
- Metal carbonyls.
- π -allyl and π -olefin complexes.
- Catalysis.
- Oxidative addition and reductive elimination.

20. Bioinorganic chemistry

Bioinorganic chemistry studies the role of inorganic elements in living systems. Metal ions are essential for enzymes, oxygen transport, electron transfer, and structural function.

Examples

- Hemoglobin and myoglobin.
- Chlorophyll.
- Cytochromes.
- Carboxypeptidase.
- Vitamin B12.

21. Solid-state chemistry

Solid-state chemistry covers crystal structures, defects, band theory, semiconductors, and lattice energy. It is important in materials science and inorganic synthesis.

22. Nuclear chemistry and radioactive elements

Many BSc inorganic syllabi also include radioactive decay, nuclear reactions, half-life, and applications of radioisotopes.

23. Analytical inorganic chemistry

This area includes qualitative and quantitative aspects of metal ions, precipitation reactions, complexometric titrations, and separation of ions.

24. Industrial and environmental applications

Inorganic chemistry has broad application in fertilizers, catalysts, ceramics, metallurgy, water treatment, pigments, battery materials, and pollution control.

25. Summary

Inorganic chemistry is the chemistry of elements, ions, and their compounds, with major emphasis on periodicity, bonding, coordination, transition metals, organometallics, bioinorganic systems, and solids. For BSc Chemistry, it is a concept-heavy and application-rich subject that forms the foundation for analytical, environmental, industrial, and materials chemistry.